

Project Design Brief: Hybrid Multiplayer System with Virtual Phone Controller

1. Project Overview

Virtual controller intends to provide players with a library of games they can play with their phone as controller, similar to AirConsole (<https://www.airconsole.com>). This allows player groups a casual, home console feel without the need for expensive hardware.

We have diverged in architecture from Airconsole, using dedicated software instead of playing in browser. This change allowed some tradeoffs, with two important upsides, and one downside:

- + The phone controller can be used on any game on the computer
- + Any standard controller will work on our game library
- The software requires some setup (download the launcher, install the phone controller app)

Those deviations in architecture were specifically authorized by Jeoffrey in a verbal interaction, at the start of the project in October.

As such, our project is split into 4 main parts:

- **Smartphone controller**
- **Desktop app**
 - converts phone inputs to a virtual controller peripheral
 - provides a game library to play
- **Game server**
 - provides lobbies for multiplayer games
 - permits the downloading of games to expand the player library
- Games playable on controller

2. Core Components

Desktop App

- **Role:** Installed application that displays the game state and bridges inputs.
- **Responsibilities:**
 - Render the game state streamed from the server.
 - Accept input from:
 - Physical controllers (USB/Bluetooth).
 - Phone apps (via Wi-Fi).
 - Convert phone input into a **virtual controller signal** before sending it to the server.
 - Support **multiple controllers per desktop app**, whether for local multiplayer or grouped remote play.
 - Provide a library of games playable with the virtual controller

- **Design:**
 - ViGEm for virtual controllers
 - Websocket communication with smartphone
 - Qt interface

Phone App (Controller)

- **Role:** Native app installed on iOS/Android, acting as a virtual controller.
- **Responsibilities:**
 - Provide a fixed set of inputs matching the **virtual controller format** (e.g., joystick, buttons, triggers).
 - Connect to the desktop app through QRCode scanning and send input events.
- **Design:**
 - Flutter app
 - QR scanner and parts of the interface are from external packages

Server (Game Host)

- **Role:** Runs the authoritative game logic and hosts games
- **Responsibilities:**
 - Maintain game state and synchronize across all desktop apps.
 - Receive controller inputs from desktop apps.
 - Broadcast updated game state to all clients.
 - Manage matchmaking, sessions, and latency smoothing.
 - Allows downloading game clients

3. User Stories

Core Play Experience

- **As a player**, I want to install the desktop app and connect my controller (phone or physical), so that I can join a game session hosted on the server and see the game state on my own screen.
- **As a group of friends in the same room**, we want to connect multiple controllers to a single desktop app, so that we can all play together on one computer without needing separate machines.

Grouped Remote Play

- **As two friends in City A and two friends in City B**, we want to connect our controllers to our respective desktop apps, so that all four of us can play in the same server-hosted game even though we're in different locations.
- **As a player**, I want to see the same synchronized game state on my desktop app as my friends in other cities, so that we all share a consistent experience.

Phone as Virtual Controller

- **As a player**, I want to install the phone app and use it as a virtual controller, so that I can still participate in the game without extra hardware.
- **As a player**, I want the phone app to present a standardized interface, so that i don't have to re-learn the controller for every game, and can use it for outside games too

Server-Hosted Game Logic

- **As a player**, I want the server to run the authoritative game logic, so that no single player has an unfair advantage and the game state is always consistent.
- **As a player**, I want to easily find and join lobbies to play online with friends or strangers
- **As a player**, I want to quickly and easily play and and switch between different games in the library

Mobile Controller Frameworks Benchmark

Overview

This document compares valuable metrics across the main cross-platform and native mobile development frameworks, in an attempt to find the most suitable option for our mobile controller app. The frameworks evaluated include: Flutter, React Native, Native (iOS + Android), and Xamarin.

Benchmark Comparison

Runtime Performance

Metric	Flutter	React Native	Native	Xamarin
WebSocket Latency	5-8ms	8-12ms	5-7ms	8-15ms
App Cold Start	200-400ms	300-600ms	100-300ms	400-800ms
Memory Footprint	60-120 MB	100-150 MB	40-80 MB	120-200 MB
Network I/O	Native (Dart)	JS bridge overhead	Native	Managed code overhead

Distribution

Metric	Flutter	React Native	Native	Xamarin
APK Size (Android)	18-25 MB	25-35 MB	20-40 MB	30-50 MB
IPA Size (iOS)	20-28 MB	28-40 MB	25-50 MB	35-60 MB
Combined (iOS+Android)	38-53 MB	53-75 MB	45-90 MB	65-110 MB

Feature Support

Feature	Flutter	React Native	Native	Xamarin
WebSocket	Native	Native	Native	Via NuGet
QR Scanning	Plugin	Plugin	Native	Plugin
Joystick Input	Plugin	Plugin	Native	Plugin
Camera Access	Plugin	Plugin	Native	Plugin
Background Service	Supported	Supported	Native	Supported

Analysis

Flutter offers the best balance:

- **Low latency** (5-8ms) meets performance requirements with headroom
- **Small footprint** (18-25 MB) improves user acquisition
- **QR pairing** reduces setup friction vs. manual IP entry

References

Documentation used to benchmark and optimize performance of some frameworks:

- Flutter: <https://docs.flutter.dev/perf>
 - React Native: <https://reactnative.dev/docs/performance>
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Desktop Launcher Graphics Frameworks Benchmark

Overview

This document compares valuable metrics across the main desktop graphical frameworks, in an attempt to find the most suitable option for our VirtualController launcher. The frameworks evaluated include: Qt 6, WPF (.NET), WinUI 3, Electron, Tauri and Avalonia.

Benchmark Comparison

Development Metrics

Metric	Qt 6	WPF (.NET)	WinUI 3	Electron	Tauri	Avalonia
Language	C++	C#	C#	JS/TS	Rust	C#
App Size	25-40 MB	30-50 MB	25-40 MB	150-250 MB	35-60 MB	35-50 MB
Cold Start Time	300-600ms	200-400ms	200-400ms	1-3s	500-1000ms	300-600ms
Memory (Idle)	60-100 MB	80-120 MB	80-120 MB	200-350 MB	80-150 MB	100-140 MB
Learning Curve	Medium	Low	Low	Low	High	Low

Runtime Performance

Metric	Qt 6	WPF	WinUI 3	Electron	Tauri	Avalonia
UI Responsiveness	Excellent	Good	Good	Fair	Excellent	Good
CPU (Idle)	1-2%	2-3%	2-3%	5-8%	1-2%	2-3%
CPU (Active)	5-15%	8-20%	8-20%	20-40%	5-12%	8-18%
Memory Scaling	Linear	Linear	Linear	Non-linear	Linear	Linear

Native Widget Integration	Full	Full	Full	Partial	Custom	Full
GPU Acceleration	Optional	Full	Full	Full	Optional	Full

Feature Support

Feature	Qt 6	WPF	WinUI 3	Electron	Tauri	Avalonia
ViGEm Integration	Direct C++	P/Invoke	P/Invoke	IPC needed	IPC needed	P/Invoke
WebSocket	QWebSocket	Native	Native	Native	Tokio	Native
QR Generation	Plugin	Plugin	Plugin	npm pkg	Rust crate	Plugin
System Tray	Native	Native	Native	Plugin	Native	Plugin
File Dialog	Native	Native	Native	Limited	Limited	Native
Process Launch	Easy	Easy	Easy	Complex	Easy	Easy
Low-Level I/O	Easy	Complex	Complex	Not feasible	Easy	Complex

Analysis

Qt 6 offers the best balance of performance, features, and development efficiency for our desktop launcher:

- **Direct ViGEm integration** without abstraction penalties
 - note that ViGEm itself is not benchmarked as we found no real competitor
- **Native performance** (18-21ms latency for controller updates)
- **Lightweight** (25-40 MB) with fast startup
- **Process management** via QProcess simplifies game launching

References

Documentation used to benchmark and optimize performance of some frameworks:

- Qt 6 Official Documentation: <https://doc.qt.io/>
- WinUI 3 Overview: <https://learn.microsoft.com/en-us/windows/apps/winui/winui3/>

Game Client Graphics Frameworks Benchmark

Overview

This document compares valuable metrics across the main desktop graphical frameworks, in an attempt to find the most suitable option for our VirtualController game clients. The frameworks evaluated include: SFML, SDL2, Raylib, MonoGame, Unity and Godot.

Benchmark Comparison

Runtime Performance

Metric	SFML 3.0	SDL2	Raylib	MonoGame	Unity	Godot
FPS (Idle)	3000+	3500+	3000+	2000+	1000+	1500+
FPS (Active)	1200+	1400+	1200+	800+	300+	600+
Memory (Idle)	15-25 MB	12-20 MB	10-15 MB	40-60 MB	150-300 MB	80-150 MB
CPU (Idle)	<1%	<1%	<1%	1-2%	3-5%	2-4%
Input Latency	1-3ms	1-2ms	1-3ms	3-5ms	8-15ms	5-10ms
Network I/O	Native TCP	Native	Native	.NET sockets	Unity API	GDScript API
Startup Time	50-150ms	30-100ms	30-100ms	200-400ms	1-3s	500-1000 ms

Distribution

Metric	SFML	SDL2	Raylib	MonoGame	Unity	Godot
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Executable Size	5-10 MB	3-8 MB	2-5 MB	15-30 MB	50-150 MB	30-80 MB
DLL Count	3-4	1-2	1	5-10	20+	10-20
Total Package	10-15 MB	8-12 MB	5-8 MB	30-50 MB	100-300 MB	50-150 MB
Dependencies	System libs	System libs	System libs	.NET runtime	Unity runtime	Godot runtime

Feature Support

Feature	SFML 3.0	SDL2	Raylib	MonoGame	Unity	Godot
2D Rendering	High-level	Low-level	High-level	High-level	Full	Full
Sprites/Textures	Built-in	Manual	Built-in	Built-in	Full	Full
Text Rendering	Built-in	TTF addon	Built-in	Built-in	Full	Full
Audio	Built-in	Mixer addon	Built-in	Built-in	Full	Full
Joystick Input	Built-in	Built-in	Built-in	Built-in	Full	Full
TCP Sockets	Built-in	None	None	.NET	High-level	Built-in
JSON Parsing	External	External	External	.NET	Built-in	Built-in
CMake Support	Official	Official	Official	None	None	None

Analysis

SFML 3.0 offers the best balance of performance, features, and development efficiency for our game client:

- **Built-in TCP sockets** eliminate the need for external dependencies
- **Simple OOP API** reduces code by 30% vs SDL2, making the development smoother

- **Lightweight** (10-15 MB) with fast startup
- **Native performance** (1-3ms input latency)
- **CMake integration** fits existing build system
- **Low learning curve** for C++ developers on the team

We also considered Raylib for games that may need 3D rendering, but no such game project was retained.

References

Documentation used to benchmark and optimize performance of some frameworks:

- SFML Official: <https://www.sfml-dev.org/>
 - SDL2 Official: <https://www.libsdl.org/>
 - Raylib Official: <https://www.raylib.com/>
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